

○ 机械运动

一、运动学的基础概念 2026-03-04

(1) Displacement, speed, acceleration, velocity

average speed: $\bar{s} = \frac{\text{total distance}}{\Delta t}$

average velocity: $\bar{v} = \frac{\Delta x}{\Delta t}$

velocity: $v = \lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t} = \frac{dx}{dt}$

acceleration: $a = \lim_{\Delta t \rightarrow 0} \frac{\Delta v}{\Delta t} = \frac{dv}{dt} = \frac{d^2x}{dt^2}$

$$\int_{v_0}^v dv = \int_{t_0}^t a(t) dt \Rightarrow v = v_0 + \int_{t_0}^t a(t) dt$$

$$\int_{x_0}^x dx = \int_{t_0}^t v(t) dt \Rightarrow a(x - x_0) = \int_{v_0}^v v(t) dv \Rightarrow v^2 - v_0^2 = 2a(x - x_0)$$

In 3D World, we can also say that- 三个分量决定一切！导数也是三个导分量组成的！举个例子： $\vec{a} = \frac{d^2x}{dt^2}\vec{i} + \frac{d^2y}{dt^2}\vec{j}$

Something interesting: $|\vec{v}| = \frac{ds}{dt}$

1. Free Fall & Throwing

$$v_y = v_{0y} - gt$$

$$\Delta t = \frac{v_{0y}}{g}$$

$$y = \frac{1}{2}gt^2 = \frac{v_{0y}^2}{2g}$$

$$\text{For throwing: } T = 2\sqrt{\frac{2h}{g}}$$

2. Circular

$$\theta = \omega t$$

$$\omega = \frac{d\theta}{dt}$$

($\vec{\omega}$: angular velocity)

$$\vec{v} = R \cos(\omega t)\vec{i} + R \sin(\omega t)\vec{j}$$

$$\vec{v} = \frac{dr}{dt} = -\omega R \sin(\omega t)\vec{i} + \omega R \cos(\omega t)\vec{j}$$

$$|\vec{v}| = \omega R$$

$$\text{Centripetal acceleration: } \vec{a} = -\omega^2 R \cos(\omega t)\vec{i} - \omega^2 R \sin(\omega t)\vec{j}$$

$$|\vec{a}| = \omega^2 R$$

3. Polar coordinate

$$\vec{e}_r = \cos \theta \vec{i} + \sin \theta \vec{j}$$

$$\vec{e}_\theta = -\sin \theta \vec{i} + \cos \theta \vec{j}$$

$$\frac{d\vec{e}_r}{d\theta} = \vec{e}_\theta$$

$$\frac{d\vec{e}_\theta}{d\theta} = -\vec{e}_r$$

$$\vec{r} = r\vec{e}_r$$

$$\frac{dr}{dt} = \frac{dr}{dt}\vec{e}_r + r \frac{d\vec{e}_r}{dt} = \frac{dr}{dt}\vec{e}_r + \vec{\omega} r \vec{e}_\theta$$

$$\vec{a} = \frac{d\vec{v}}{dt} = \frac{d}{dt} \left(\frac{dr}{dt}\vec{e}_r + r \frac{d\vec{e}_r}{dt} \right) = (\text{after a long process}) \left(\frac{d^2r}{dt^2} - \omega^2 r \right) \vec{e}_r + \left(r \frac{d\omega}{dt} + 2\omega \frac{dr}{dt} \right) \vec{e}_\theta$$